Blinded Visual Question-Answering Evaluation of Qwen3-VL Plus on Gear Assemblies

Comprehensive methodology, performance, error, token, and cost report

Field	Value
Model	qwen3-vl-plus
Evaluation date	2026-08-11
Repository commit	360772e84d93f63010c806fc0faf21654bb8c4fc
Scope	33 top-level base packages; hidden content excluded

Purpose: document methodology, exact-answer performance, question-type error rates, token use, estimated list-price cost, and per-item failures.

Prepared from retained provider responses, blind grading outputs, and independent reconciliation checks.

Abstract

This evaluation measured exact-answer performance of qwen3-vl-plus on 33 top-level base packages from the Gear Assemblies QA repository. Each request contained one question and six rendered CAD views. Hidden content, dot-directories, ground truth, and unrelated repository files were excluded from inference. Predictions were graded through a separate blind handoff. The model answered 5 of 33 items correctly, yielding 15.15% accuracy (95% Wilson CI, 6.7-30.9%) and an 84.85% error rate. Component-count accuracy was 50.00%; transmission ratios, tooth counts, and gear type were 0.00%. The run used 136,043 tokens. Published US or Global list rates imply an estimated cost of USD 0.019766571, or about 1.98 cents.

15.15%	84.85%	5 / 33	1.98 cents
Accuracy	Error rate	Exact answers	List-price estimate

1. Introduction

Mechanical visual question answering combines recognition, counting, cross-view correspondence, geometric interpretation, and kinematic reasoning. A model may recognize a gear assembly yet fail to recover an exact tooth count or transmission ratio. This study measured those capabilities without exposing CAD metadata, hidden variants, or the answer key.

The study used qwen3-vl-plus, the documented Qwen3 vision-language Plus alias served through Alibaba Cloud Model Studio's OpenAI-compatible interface [1]. The requested label Qwen 3.8 Plus did not match a published vision model identifier, so the documented vision-capable alias was selected. The objective was to quantify exact-answer accuracy, characterize errors by question type, and account for token and list-price cost.

2. Methods

2.1 Dataset and scope

The source was axioforgeai/gear-assemblies-qa at commit 360772e84d93f63010c806fc0faf21654bb8c4fc [4]. A shallow selector admitted only top-level directories with ModelNN_QANN identifiers, producing 33 base packages. Every package contributed one prompt and six PNG views: isometric front-right-top, isometric front-left-top, isometric rear-right-top, front, top, and left. The evaluated set contained 33 prompts and 198 images.

The staging procedure did not descend into the hidden directory and rejected hidden, dot-prefixed, absolute, empty, and traversal paths. Ground-truth CSV files and repository-level review files were not copied into inference staging.

2.2 Model request

Each package was submitted independently to https://dashscope-us.aliyuncs.com/compatible-mode/v1, an Alibaba Cloud US endpoint [3]. Requests used temperature 0, max_tokens 128, non-streaming output, and thinking disabled. The original question was followed by six image data URLs. The exact system instruction was:

> You are completing a blinded visual question-answering benchmark on a complete CAD gear assembly. Use only the question and six images in this request. Return exactly one JSON object with one key named answer. The value must be the shortest direct answer: a number only for numeric questions, or the requested category or term only. Do not include reasoning, units, markdown, or extra keys.

All 33 retained responses ended with finish reason stop. No empty or failed answer entered the scored set.

2.3 Blind separation and independent audit

Two sequential subagents received non-overlapping handoffs without inherited conversation context. The inference-side agent could inspect staged base prompts, images, configuration, and outputs but could not inspect answers, grading files, hidden directories, or dot-directories. Because this isolated agent did not carry the user's authorization for a billable network call, the transcript-bearing coordinator executed the fixed runner against the same staged inputs. The agent then audited model identity, completeness, failures, tokens, scope, and credential hygiene.

The grading agent was launched after inference and received only normalized predictions and a separate base-answer handoff. It could not inspect questions, images, raw responses, usage, the API key, source repository, or inference workspace. The coordinator independently matched all 33 predictions and answers and recomputed every decision. This was procedural blindness on a shared filesystem, not an operating-system sandbox.

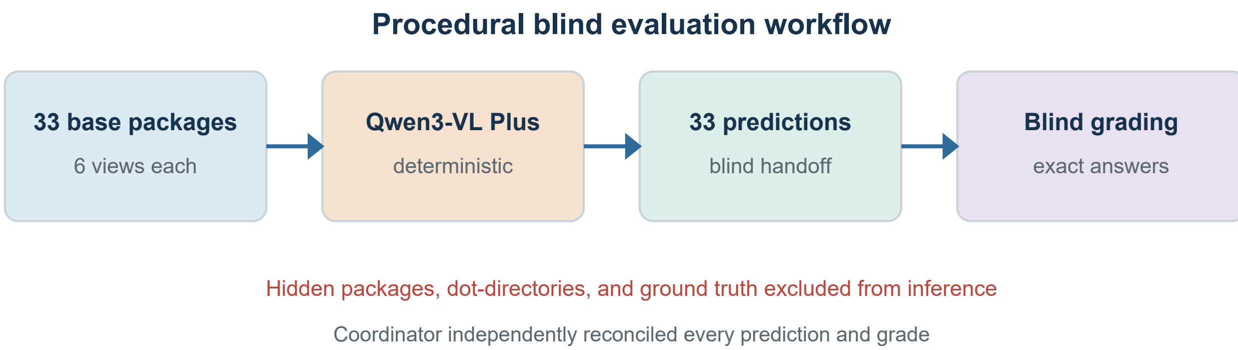

Figure 1. Evaluation workflow. Hidden content and ground truth were excluded from inference; grading used separated answer strings.

2.4 Scoring and descriptive analysis

Numeric ground truth required one parsed numeric token equal to the expected value within absolute tolerance 1e-9. Discrete text was Unicode-normalized, stripped of surrounding whitespace and punctuation, lowercased, and matched exactly. Missing, ambiguous, unparsable, and failed predictions were defined as incorrect before grading.

Accuracy was correct answers divided by the relevant denominator; error rate was one minus accuracy. Wilson 95% confidence intervals describe binomial uncertainty. A post hoc descriptive taxonomy classified questions as transmission ratio, gear tooth count, component count, axis angle, or gear type. Mean absolute error, median absolute error, and mean absolute percentage error were calculated within numeric categories.

2.5 Token and cost accounting

Token counts were summed from provider usage objects. Alibaba Cloud lists qwen3-vl-plus in the US or Global deployment at USD 0.143 per million input tokens and USD 1.434 per million output tokens for requests no larger than 32K input tokens [2]. The largest request contained 4,183 prompt tokens, so every request stayed in that tier. The calculation is a published list-price estimate, not a verified account invoice; credits, discounts, taxes, provider rounding, or account adjustments may differ.

3. Results

3.1 Aggregate and category performance

The model returned 5 correct and 28 incorrect answers. Overall accuracy was 15.15% (95% Wilson CI, 6.7-30.9%), and error rate was 84.85%. Numeric accuracy was 15.62% (5/32); the single discrete item was incorrect.

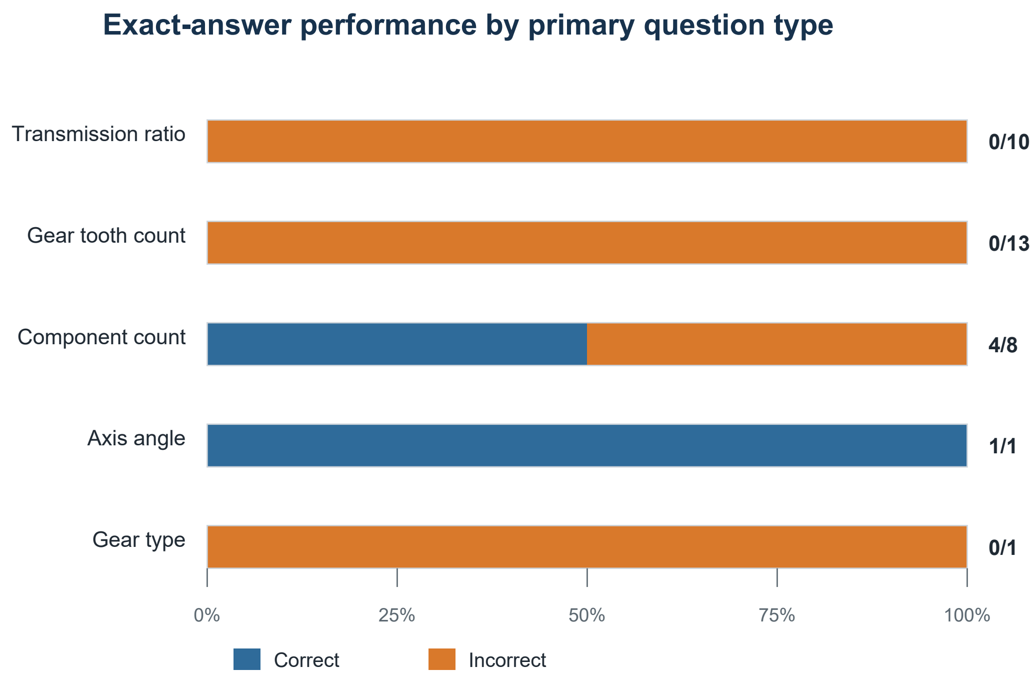

Figure 2. Exact-answer performance by primary question type. Right-side labels show correct answers divided by category size.

Question type	n	Correct	Accuracy	Error	95% CI	MAE	MAPE
Transmission ratio	10	0	0.00%	100.00%	0.0-27.8%	7.735	53.72%
Gear tooth count	13	0	0.00%	100.00%	0.0-22.8%	18.077	38.15%
Component count	8	4	50.00%	50.00%	21.5-78.5%	0.875	26.04%
Axis angle	1	1	100.00%	0.00%	20.7-100.0%	0.000	0.00%
Gear type	1	0	0.00%	100.00%	0.0-79.3%	n/a	n/a

Table 1. Accuracy, error rate, binomial uncertainty, and within-category numeric error magnitude.

Transmission-ratio and tooth-count categories both had 100% error rates. Component counting produced four correct answers in eight trials. The singleton axis-angle item was correct, and the singleton gear-type item was incorrect. Confidence intervals are wide for small categories.

3.2 Subtype analysis

Question subtype	n	Correct	Accuracy	Error rate
Worm train ratio	1	0	0.00%	100.00%
Planetary ratio	2	0	0.00%	100.00%
Other train or gearbox ratio	7	0	0.00%	100.00%
Ring gear teeth	3	0	0.00%	100.00%
Sun gear teeth	3	0	0.00%	100.00%
Planet gear teeth	1	0	0.00%	100.00%
Smaller gear teeth	3	0	0.00%	100.00%
Larger gear teeth	3	0	0.00%	100.00%
Planet-gear quantity	3	3	100.00%	0.00%
Assembly inventory count	5	1	20.00%	80.00%
Axis angle	1	1	100.00%	0.00%
Gear-shape classification	1	0	0.00%	100.00%

Table 2. Descriptive performance by question subtype.

All three planet-gear quantity questions were correct, whereas only one of five assembly inventory questions was correct. Every tested ratio mechanism and every tooth-count target had a 100% error rate.

3.3 Numeric error direction

Of 32 numeric predictions, 21 were underestimates, 6 were overestimates, and 5 were exact. Among the 27 numeric errors alone, 77.78% were underestimates and 22.22% were overestimates.

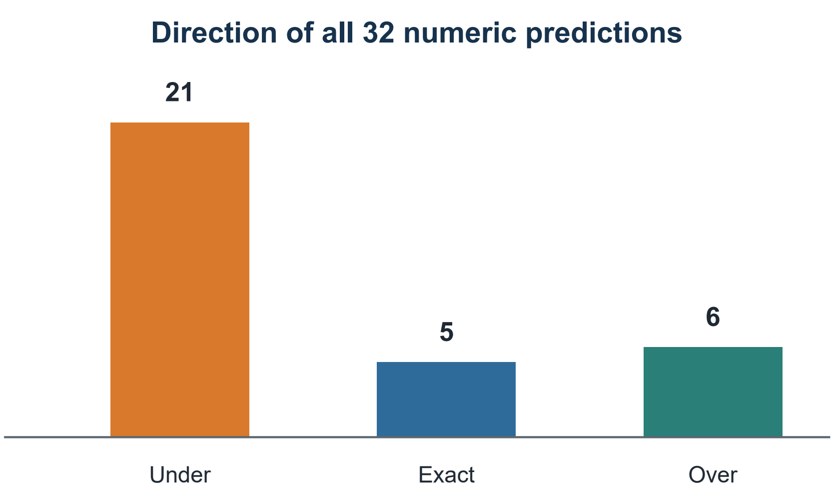

Figure 3. Direction of numeric predictions relative to ground truth.

Transmission-ratio questions had MAE 7.735 and MAPE 53.72%. Tooth-count questions had MAE 18.077 teeth and MAPE 38.15%. Component counts had MAE 0.875 and MAPE 26.04%.

3.4 Token use, response span, and estimated cost

Measure	Observed value
Successful requests	33
Prompt tokens	135,801
Image tokens	132,396
Text prompt tokens	3,405
Cached tokens	0
Completion tokens	242
Total tokens	136,043
Prompt tokens per request	3,568 min; 4115.18 mean; 4,183 max
Completion tokens per request	3 min; 7.33 mean; 9 max
Input list-price estimate	USD 0.019419543
Output list-price estimate	USD 0.000347028
Total list-price estimate	USD 0.019766571 (about 1.98 cents)
Estimated cost per question	USD 0.000598987
Estimated cost per correct answer	USD 0.003953314

Table 3. Provider-reported token usage and published list-price estimate.

The 33 saved completions spanned 109.8 seconds, or 1.83 minutes. This interval reflects sequential API response collection and excludes staging, auditing, grading, and reporting.

3.5 Correct-answer control cases

Package	Type	Question	Correct	Predicted
Model01_QA02	Axis angle	What is the angle between the axes of the worm and the worm wheel?	90	90
Model02_QA03	Component count	How many distinct parts are there in this assembly?	5	5
Model03_QA04	Component count	How many planet gears are there in this planetary gear train?	3	3
Model04_QA05	Component count	How many planet gears are there in this planetary gear train?	3	3
Model09_QA04	Component count	How many planet gears are there?	4	4

Table 4. Five correct items retained as controls for response extraction and grading.

Four exact cases involved direct visible counts or a direct spatial relation. None required tooth enumeration or transmission-ratio calculation.

4. Discussion

The two largest mechanically demanding categories failed completely. None of the 10 transmission-ratio questions and none of the 13 tooth-count questions was correct. These 23 items drive the low aggregate score and indicate that six rendered views were insufficient for the tested model and prompt to recover exact tooth counts or translate visible structure into reliable kinematic ratios.

Component counting was less uniformly difficult. Planet gears were counted correctly in all three planetary assemblies, suggesting that salient repeated objects can be recognized when boundaries are clear. Broader inventory questions were much weaker: only one of five was correct. Supports, bearings, gears, and shafts may be occluded, duplicated across views, or difficult to distinguish as unique parts.

Numeric errors were asymmetric: 21 of 27 were underestimates. Several tooth predictions repeated plausible but incorrect lower values such as 12, 24, and 72. This does not reveal the model's internal mechanism, but it is consistent with coarse visual estimation rather than exact enumeration.

The axis-angle success and gear-type failure each represent one observation, so neither supports a stable category-level conclusion. Wide Wilson intervals make this small-sample uncertainty explicit.

5. Limitations and recommended follow-up

This benchmark covers one model alias, one deterministic run, one prompt, one repository commit, and only the 33 base packages. Hidden variants were intentionally excluded. The taxonomy was post hoc, exact-match grading awarded no partial credit, and reasoning was suppressed. An incorrect ratio therefore cannot be decomposed into perception, enumeration, kinematic setup, arithmetic, or output-selection errors.

A stronger follow-up would pin a dated model snapshot, repeat each item, and compare six-view input with targeted close-ups or view ablations. A tool-assisted condition could let the model request crops, count repeated tooth edges, and execute a transparent ratio calculation from extracted values.

6. Conclusion

Under the tested base-only six-view protocol, qwen3-vl-plus achieved 15.15% exact-answer accuracy and an 84.85% error rate. It succeeded on the three planet-gear quantity questions and two isolated simple cases, but every transmission-ratio and tooth-count item was incorrect. Most numeric errors were underestimates. These findings apply to this alias, prompt, date, and source commit.

Data availability and audit trail

The local evaluation directory retains staged base prompts and views, raw provider responses, normalized predictions, run state, blind grading handoffs, and per-item grading output. Independent reconciliation matched all 33 predictions and answers, recomputed every decision, confirmed 33 raw responses, and found no hidden or dot-prefixed staged path. The source Git checkout remained clean.

Security and operational note

The API-key file remained outside the Git checkout. Initial local preflight incorrectly treated two credential lines as one header value, and a local header-validation diagnostic briefly duplicated the text into a failure JSON. Validation stopped before an API request was sent. The file was deleted, linewise parsing and redaction were added, and retained outputs passed a credential scan. Credential rotation remains recommended. TLS verification stayed enabled through Node.js system-CA support.

References

[1] Alibaba Cloud. Qwen-VL compatible with OpenAI. Accessed Aug. 11, 2026. https://www.alibabacloud.com/help/en/model-studio/qwen-vl-compatible-with-openai

[2] Alibaba Cloud. Model Studio model pricing. Accessed Aug. 11, 2026. https://www.alibabacloud.com/help/en/model-studio/model-pricing

[3] Alibaba Cloud. API endpoint. Accessed Aug. 11, 2026. https://www.alibabacloud.com/help/en/model-studio/base-url

[4] axioforgeai. gear-assemblies-qa. GitHub commit 360772e84d93f63010c806fc0faf21654bb8c4fc. https://github.com/axioforgeai/gear-assemblies-qa/tree/360772e84d93f63010c806fc0faf21654bb8c4fc